CLIMB at VBS 2027: Multi-Signal Video Retrieval with Team-Based Collaboration

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Abstract. We present CLIMB, an interactive video retrieval system built for the Video Browser Showdown 2027. CLIMB retrieves information across four complementary signals: visual embeddings, on-screen text, generated captions, and speech transcripts. By combining rank-based fusion with a hybrid of lexical and semantic matching, the system ensures broad retrieval coverage. Additionally, queries can be chained into temporal sequences, providing increased flexibility while sharply narrowing the result set. We treat retrieval as a shared team activity, allowing operators to send hints to each other and automatically hiding teammate-submitted scenes for ad-hoc video search. Crucially, core functions such as search, browsing, and submission remain fully operational even if the shared collaboration service becomes unreachable.

Keywords: Video Browser Showdown · Interactive Video Retrieval · Collaborative Search · Semantic Embedding.

1 Introduction

The Video Browser Showdown (VBS) [5, 6, 11] is an interactive video retrieval competition that asks participants to solve known-item search (KIS), ad-hoc video search (AVS), and visual question-answering (VQA/QAS) tasks against large-scale video collections. For VBS 2027, the main collection is V3C [12] with 28,450 videos, roughly 3,800 hours, and about 8.7 TB of full-resolution video data, along with the smaller collections MVK [17] and GynSurgLHE (LHE) [8].

Successfully completing these tasks presents multiple fundamental challenges. A single visual embedding is often too vague to capture precise task hints, such as specific on-screen text or spoken dialog, and for difficult tasks, teammates have no direct way to point each other to promising results. Additionally, because AVS scoring rewards distinct correct shots, a team of n operators searching independently spends a substantial share of its submissions re-finding what a teammate has already sent.

CLIMB addresses all of them. It indexes every video as scenes and keyframes, retrieves over four independent signals, fuses them by rank, allows for operators to send clues to their teammates, and treats an AVS task as a single shared activity, vanishing duplicate scenes from everybody's list.

Concretely, we contribute:

1. Keyframe-level retrieval fusing four heterogeneous signals by rank (Sect. 4.1).
2. Temporal chaining, `A >> B >>(120) C`, with a full query per stage (Sect. 4.3).
3. Network-resilient team collaboration: shared AVS state and hints (Sect. 6).
4. An index engineered for commodity hardware: binary-quantized HNSW with oversample-and-rerank (Sect. 3).

2 System Overview

CLIMB is built from six components (see Fig. 1). Decoding each video, segmenting it into scenes, selecting keyframes, computing embeddings, and text recognition in frames via OCR, as well as captioning and transcribing, happens in an offline pipeline described in Sect. 3. The resulting scenes, keyframes, and four indexes that the retrievers query land in a PostgreSQL database.

At query time, the system runs as two processes, a Node.js backend transmitting media to the frontend, handling DRES [10, 14] communication and collaboration features (Sect. 6), as well as a Python engine handling models, query embedding, and rank fusion. Both connect to the database: the backend reads metadata directly, delegating only search execution to Python. The frontend (Sect. 5) communicates exclusively with the Node.js backend.

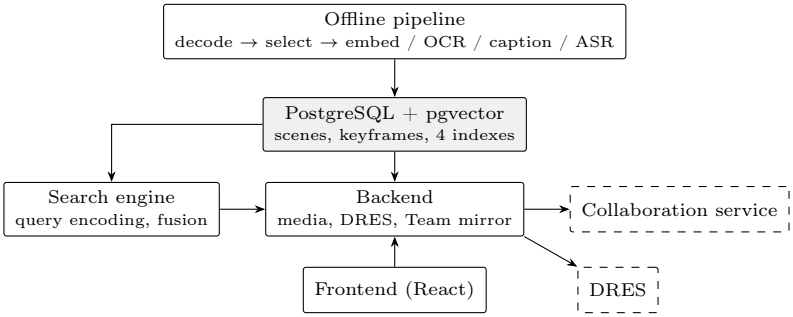


Fig. 1. CLIMB architecture. All components can run locally on each machine, except for two external services: DRES and the collaboration service, which is deployed once.

3 Indexing Pipeline

3.1 Scenes and Keyframe Selection

The first step of the indexing pipeline is shot segmentation and keyframe extraction. For all three datasets, scene boundaries are shipped with the dataset. The segmentation has been done with TransNetv2 [16] for V3C and a specific keyframe extraction algorithm [15], based on ORB keypoint tracking [13] for the LHE [8] and MVK videos [17]. MVK clips from the same dive are shipped as

separate clips, but they are grouped into one recording on a shared timeline, so exclusion, temporal chains and AVS coverage work per dive, while DRES submissions still reference the individual clips.

Within each scene, candidate frames are decoded at 2 fps, and the scene receives one keyframe per four seconds of duration, at least two and at most 32. Fades and blank frames (very dark, very bright, or low-contrast) are discarded first, unless they contain edges, so that title cards and credits remain available for OCR. Each remaining candidate is described by a cheap, model-free fingerprint: a colour histogram, a perceptual hash, and the edge density. Selection follows a greedy farthest-point strategy [3], seeded with the frame closest to the scene's mean fingerprint.

3.2 Feature Extraction

CLIMB extracts four signals, two per keyframe and two per scene. Each keyframe is embedded into a 1024-dimensional space by SigLIP2 [18] and searched for on-screen text by PP-OCR [21]. Since surgical footage lies far from SigLIP2's training data, LHE keyframes are additionally embedded with PeskaVLP [20].

The two scene signals are textual and embedded into a 1024-dimensional space by multilingual-e5-large [19]. Given its cost and intra-scene redundancy, Qwen3-VL-4B-Instruct [1] captions only each scene's first keyframe. Audio transcription is handled by Whisper large-v3 [9]. Transcriptions produced from music or noise, identified as repetitive or isolated from other speech, are excluded.

Embeddings are indexed with HNSW [7] over binary-quantized vectors, and candidates are reranked against their half-precision vectors. Quantization cuts the footprint for 12.4M keyframes from 28 GB at half precision to 5.3 GB.

4 Retrieval Model

4.1 Rank-Based Fusion at Keyframe Level

As discussed above, we use four different encoders to encode as broad a spectrum of information as possible. In order to avoid sensitive and model-specific calibration, we use a rank-based fusion approach, explained below.

Each signal returns a list of ranked objects. While the visual and OCR signal natively return a list of ranked keyframes, the caption and audio signals are scene-based. Thus, each keyframe inherits the rank of its parent scene. The keyframes are then merged via weighted Reciprocal Rank Fusion (RRF) [2]. This works similarly to normal RRF but weighs the signals depending on their source, allowing us to highlight keyframes found via specific encoders. The fused ranking score for a keyframe $d \in D$, over a set of encoders R, is defined as:

$$RRFscore(d \in D) = \sum_{r \in R} \frac{w(r)}{60 + r(d)} \tag{1}$$

Where $r(d)$ is the rank assigned by encoder $r \in R$ with weight $w(r)$. The constant 60 was emprically determined in [2] and reduces the dominance of top-ranked keyframes. For V3C, the uploader's video description and tags are additionally matched against the query as a video-level prior, adding an additional RRF term for each already retrieved keyframe's video's rank, re-ordering the results. The system's default weights are 1 for visual, caption, speech and metadata, 4 for OCR and 8 for explicit matches via the `said:` or `text:` operator (Sect. 4.3).

4.2 Semantic and Lexical matching

Although cross-modal visual retrieval with SigLIP2 is inherently semantic, the remaining text-to-text signals could rely on lexical matching. For generated captions, however, this fails on vocabulary mismatch: a query for "law enforcement" cannot match a caption describing "a police uniform", and neither "law" nor "enforcement" appears in any of our captions.

ASR transcripts are also embedded semantically to allow for conceptual and cross-lingual search, but users can invoke a *said:* operator for exact lexical matching of specific quotes. In contrast, OCR operates as a purely lexical signal to pinpoint exact on-screen text provided in task hints.

4.3 Query Language and Temporal Chains

A plain text query searches all four signals at once, and the operator prefixes in Table 1 narrow it down. Each restricts the search to the videos that satisfy it, ranking the plain text only within them. Once the prefixes leave at most 300 videos, they are searched exhaustively, bypassing the vector index.

For tasks describing an ordered sequence of scenes, `>>` joins queries into a temporal chain. Each stage is a complete query including its own prefixes, so `text: pegasus >> mountain bike race` finds a scene showing the text "pegasus", followed by one that matches a mountain bike race. Consecutive stages must lie in the same video and are per default at most 60 seconds apart. Writing `>>(120)` raises this to 120 seconds, up to ten minutes. A query has at most four stages.

Each stage returns a ranked list of keyframes, grouped by video. Videos lacking a keyframe for any stage are excluded. For every first-stage hit, the best continuation is found by working backwards from the last stage, so each pair of adjacent stages is compared only once. At most three chains are kept per video, and all chains are ranked by their score, the sum of their stages' RRF scores.

5 Interactive Search Interface

The CLIMB user interface is shown in Fig. 2. A search typically follows an iterative loop. The operator selects a collection (V3C, MVK, or LHE) and enters a query, which returns a grid of keyframes. Selecting one plays a five-second scene preview from its timestamp, while the filmstrip below the player shows all keyframes for quick navigation. The submission time is prefilled from the

Table 1. Query syntax. Prefixes are video-level requirements. They are repeatable and can be combined. `>>` chains queries temporally within a single video.

Input	Effect
`a drawing of people on a beach`	Global search (all signals)
`text: state capitol`	Scenes showing this text
`said: new york`	Scenes in which this is said
`tag: travel`	Only videos tagged with `travel`
`video:00191 video:00192`	Only these videos
`-video:00191`	Exclude this video
`a bridge >> beach >> sunset`	Sequential search (same video)

keyframe but stays editable for fine-tuning. *Submit to DRES* sends the selection to the evaluation server, *Exclude video from search* removes a video from the results, and when textual queries fall short, *Find Similar* enables visual query-by-example. For VQA tasks, operators type the answer below the filmstrip and submit it as text alone or together with the selected shot. Results are cached, so scrolling loads further pages without re-running the search.

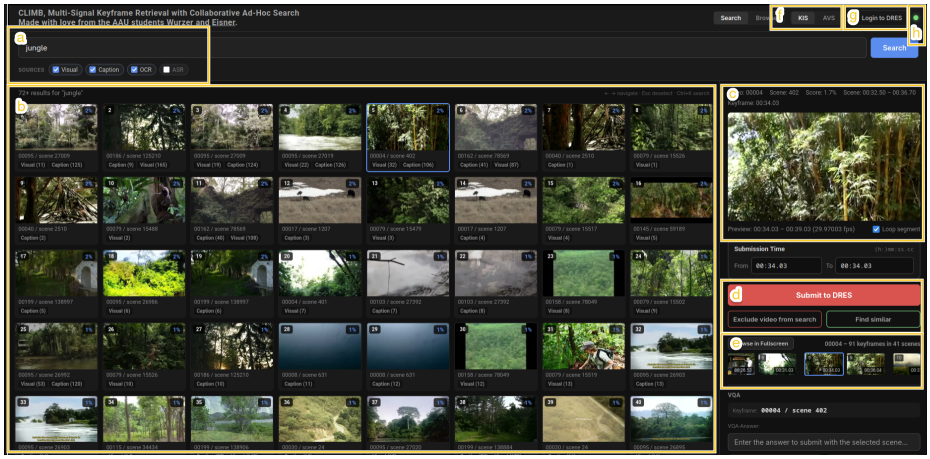


Fig. 2. The CLIMB interface: (a) query bar and source filters, (b) dataset selection, (c) scene result grid, (d) detail panel and video player, (e) submission and search-refinement controls, (f) keyframe filmstrip, (g) AVS session tab, (h) DRES connection controls, (i) Team connection controls, and (j) system health status.

6 Team Collaboration

At VBS, every operator runs a complete local CLIMB instance, avoiding network dependency for media delivery. Nonetheless, allowing operators to work together

is highly beneficial, as teammates can share promising leads, reducing wrong or duplicate submissions while speeding up the search.

To achieve this, operators can send a teammate a hint: a keyframe or an entire video, along with an optional note. The hint appears as a notification, which when clicked opens a browsing view of all keyframes with the sent keyframe selected. Whether opened or dismissed, it is kept in an inbox, so that operators can come back to it later.

For AVS, collaboration goes further. Because AVS rewards distinct correct shots, a team in CLIMB shares one AVS session. Submitted scenes are hidden from the results of all team members. Other scenes of these videos are marked as covered, steering the team towards new videos, but stay visible, since one video can hold several correct shots.

This shared state is maintained by a lightweight collaboration service, deployed once. Operators join a team by name and password, with each backend keeping a local copy of its team's state (the current AVS session and all hints), refreshed every four seconds. Importantly, searches and submissions must not depend on the speed or availability of the collaboration service. Searches therefore read only the local copy, and submissions go to DRES first and are recorded with the service afterwards, so a slow service never delays them. If the service becomes unreachable, the local copy is kept rather than cleared, so the team's exclusions survive an unstable connection.

7 Preliminary Results

A far simpler CLIMB prototype won our university's course-internal retrieval competition, but VBS is a much greater challenge. CLIMB is implemented for the full V3C collection, yet all metrics below were measured on a 200-video subset (26 h, 14,345 scenes, 35,558 keyframes).

Compared with exhaustive exact search (73 ms), oversampling 1,000 candidates before exact reranking recovers on average 95.6% of the true top 20 in 17 ms, 2,000 candidates recovers 98.3% in 33 ms, and 4,000 99.4% in 63 ms. CLIMB defaults to 2,000 candidates, while keeping this parameter configurable.

Since neither MVK nor LHE resembles the imagery SigLIP2 was trained on, we tested domain-specific models on both. Fusing PeskaVLP with SigLIP2 raises the relevant share of the first LHE result page from 46% to 54%, whereas BioCLIP 2.5 [4] alone lowers the share of MVK queries with a relevant frame on the first page from 93% to 64%, so it is not used.

8 Conclusion

In this paper, we present CLIMB, an interactive video retrieval system for VBS 2027. It uses rank-based fusion to combine visual embeddings, on-screen text, generated captions, and transcripts at the keyframe level and supports temporally chained queries. A collaborative mode hides teammate-submitted scenes and degrades gracefully, keeping core search functions fully available offline.

References

1. Bai, S., Cai, Y., Chen, R., Chen, K., Chen, X., Cheng, Z., Deng, L., Ding, W., Gao, C., Ge, C., et al.: Qwen3-VL technical report. arXiv preprint arXiv:2511.21631 (2025)
2. Cormack, G.V., Clarke, C.L.A., Buettcher, S.: Reciprocal rank fusion outperforms Condorcet and individual rank learning methods. In: Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval. p. 758–759. SIGIR '09, Association for Computing Machinery, New York, NY, USA (2009). https://doi.org/10.1145/1571941.1572114, https://doi.org/10.1145/1571941.1572114
3. Gonzalez, T.F.: Clustering to minimize the maximum intercluster distance. Theoretical Computer Science **38**, 293–306 (1985). https://doi.org/https://doi.org/10.1016/0304-3975(85)90224-5, https://www.sciencedirect.com/science/article/pii/0304397585902245
4. Gu, J., Stevens, S., Campolongo, E., Thompson, M., Zhang, N., Wu, J., Kopanev, A., Mai, Z., White, A., Balhoff, J., et al.: BioCLIP 2: Emergent properties from scaling hierarchical contrastive learning. Advances in Neural Information Processing Systems **38**, 102778–102811 (2025)
5. Lokoč, J., Andreadis, S., Bailer, W., Duane, A., Gurrin, C., Ma, Z., Messina, N., Nguyen, T.N., Peška, L., Rossetto, L., et al.: Interactive video retrieval in the age of effective joint embedding deep models: lessons from the 11th VBS. Multimedia Systems pp. 1–24 (2023)
6. Lokoč, J., Bailer, W., Schoeffmann, K., Muenzer, B., Awad, G.: On influential trends in interactive video retrieval: Video Browser Showdown 2015–2017. IEEE Transactions on Multimedia **20**(12), 3361–3376 (2018). https://doi.org/10.1109/TMM.2018.2830110
7. Malkov, Y.A., Yashunin, D.A.: Efficient and robust approximate nearest neighbor search using hierarchical navigable small world graphs. IEEE Transactions on Pattern Analysis and Machine Intelligence **42**(4), 824–836 (2020). https://doi.org/10.1109/TPAMI.2018.2889473
8. Nasirihaghighi, S., Ghamsarian, N., Peschek, L., Munari, M., Husslein, H., Sznitman, R., Schoeffmann, K.: GynSurg: A comprehensive gynecology laparoscopic surgery dataset. In: Proceedings of the 33rd ACM International Conference on Multimedia. p. 13141–13147. MM '25, Association for Computing Machinery, New York, NY, USA (2025). https://doi.org/10.1145/3746027.3758267, https://doi.org/10.1145/3746027.3758267
9. Radford, A., Kim, J.W., Xu, T., Brockman, G., Mcleavey, C., Sutskever, I.: Robust speech recognition via large-scale weak supervision. In: Krause, A., Brunskill, E., Cho, K., Engelhardt, B., Sabato, S., Scarlett, J. (eds.) Proceedings of the 40th International Conference on Machine Learning. Proceedings of Machine Learning Research, vol. 202, pp. 28492–28518. PMLR (23–29 Jul 2023), https://proceedings.mlr.press/v202/radford23a.html
10. Rossetto, L., Gasser, R., Sauter, L., Bernstein, A., Schuldt, H.: A system for interactive multimedia retrieval evaluations. In: Lokoč, J., Skopal, T., Schoeffmann, K., Mezaris, V., Li, X., Vrochidis, S., Patras, I. (eds.) MultiMedia Modeling. pp. 385–390. Springer International Publishing, Cham (2021)
11. Rossetto, L., Schoeffmann, K., Gurrin, C., Lokoc, J., Bailer, W.: Results of the 2025 Video Browser Showdown. ArXiv **abs/2509.12000** (2025), https://doi.org/10.48550/arXiv.2509.12000

12. Rossetto, L., Schuldt, H., Awad, G., Butt, A.A.: V3C – a research video collection. In: Kompatsiaris, I., Huet, B., Mezaris, V., Gurrin, C., Cheng, W.H., Vrochidis, S. (eds.) MultiMedia Modeling. pp. 349–360. Springer International Publishing, Cham (2019)
13. Rublee, E., Rabaud, V., Konolige, K., Bradski, G.: ORB: An efficient alternative to SIFT or SURF. In: 2011 International Conference on Computer Vision. pp. 2564–2571 (2011). https://doi.org/10.1109/ICCV.2011.6126544
14. Sauter, L., Gasser, R., Schuldt, H., Bernstein, A., Rossetto, L.: Performance evaluation in multimedia retrieval. ACM Transactions on Multimedia Computing, Communications and Applications (Oct 2024). https://doi.org/10.1145/3678881, https://doi.org/10.1145/3678881
15. Schoeffmann, K., Del Fabro, M., Szkaliczki, T., Böszörmenyi, L., Keckstein, J.: Keyframe extraction in endoscopic video. Multimedia Tools and Applications **74**, 11187–11206 (2015)
16. Souček, T., Lokoč, J.: Transnet V2: an effective deep network architecture for fast shot transition detection. arXiv preprint arXiv:2008.04838 (2020)
17. Truong, Q.T., Vu, T.A., Ha, T.S., Lokoč, J., Wong, Y.H., Joneja, A., Yeung, S.K.: Marine Video Kit: a new marine video dataset for content-based analysis and retrieval. In: International conference on multimedia modeling. pp. 539–550. Springer (2023)
18. Tschannen, M., Gritsenko, A., Wang, X., Naeem, M.F., Alabdulmohsin, I.M., Parthasarathy, N., Evans, T., Beyer, L., Xia, Y., Mustafa, B., Hénaff, O., Harmsen, J., Steiner, A., Zhai, X.Q.: SigLIP 2: Multilingual vision-language encoders with improved semantic understanding, localization, and dense features. ArXiv **abs/2502.14786** (2025), https://arxiv.org/abs/2502.14786
19. Wang, L., Yang, N., Huang, X., Yang, L., Majumder, R., Wei, F.: Multilingual E5 text embeddings: A technical report. arXiv preprint arXiv:2402.05672 (2024)
20. Yuan, K., Srivastav, V.K., Navab, N., Padoy, N.: Procedure-aware surgical video-language pretraining with hierarchical knowledge augmentation. ArXiv **abs/2410.00263** (2024), https://api.semanticscholar.org/CorpusID:273022518
21. Zhang, Y.B., Wang, X., Lin, M.H., Zhang, Y., Deng, P., Sun, T., Gao, T., Zhang, Z.L., Liu, J.X., Zhou, C., Liu, H., Liang, S., Cui, C., Liu, Y., Yu, D.H., Ma, Y.J.: PP-OCRv6: From 1.5m to 34.5m parameters, surpassing billion-scale vlms on ocr tasks. ArXiv **abs/2606.13108** (2026)